

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear functions	Hard

Question

$f(x) = 4x^2 - 50x + 126$

The given equation defines the function f . For what value of x does $f(x)$ reach its minimum?

Correct Answer: 25/4, 6.25

Rationale

The correct answer is $\frac{25}{4}$. The given equation can be rewritten in the form $f(x) = a(x - h)^2 + k$, where a , h , and k are constants. When $a > 0$, h is the value of x for which $f(x)$ reaches its minimum. The given equation can be rewritten as $f(x) = 4(x^2 - \frac{50}{4}x) + 126$, which is equivalent to $f(x) = 4(x^2 - \frac{50}{4}x + (\frac{50}{8})^2 - (\frac{50}{8})^2) + 126$. This equation can be rewritten as $f(x) = 4((x - \frac{50}{8})^2 - (\frac{50}{8})^2) + 126$, or $f(x) = 4(x - \frac{50}{8})^2 - 4(\frac{50}{8})^2 + 126$, which is equivalent to $f(x) = 4(x - \frac{25}{4})^2 - \frac{121}{4}$. Therefore, $h = \frac{25}{4}$, so the value of x for which $f(x)$ reaches its minimum is $\frac{25}{4}$. Note that 25/4 and 6.25 are examples of ways to enter a correct answer.